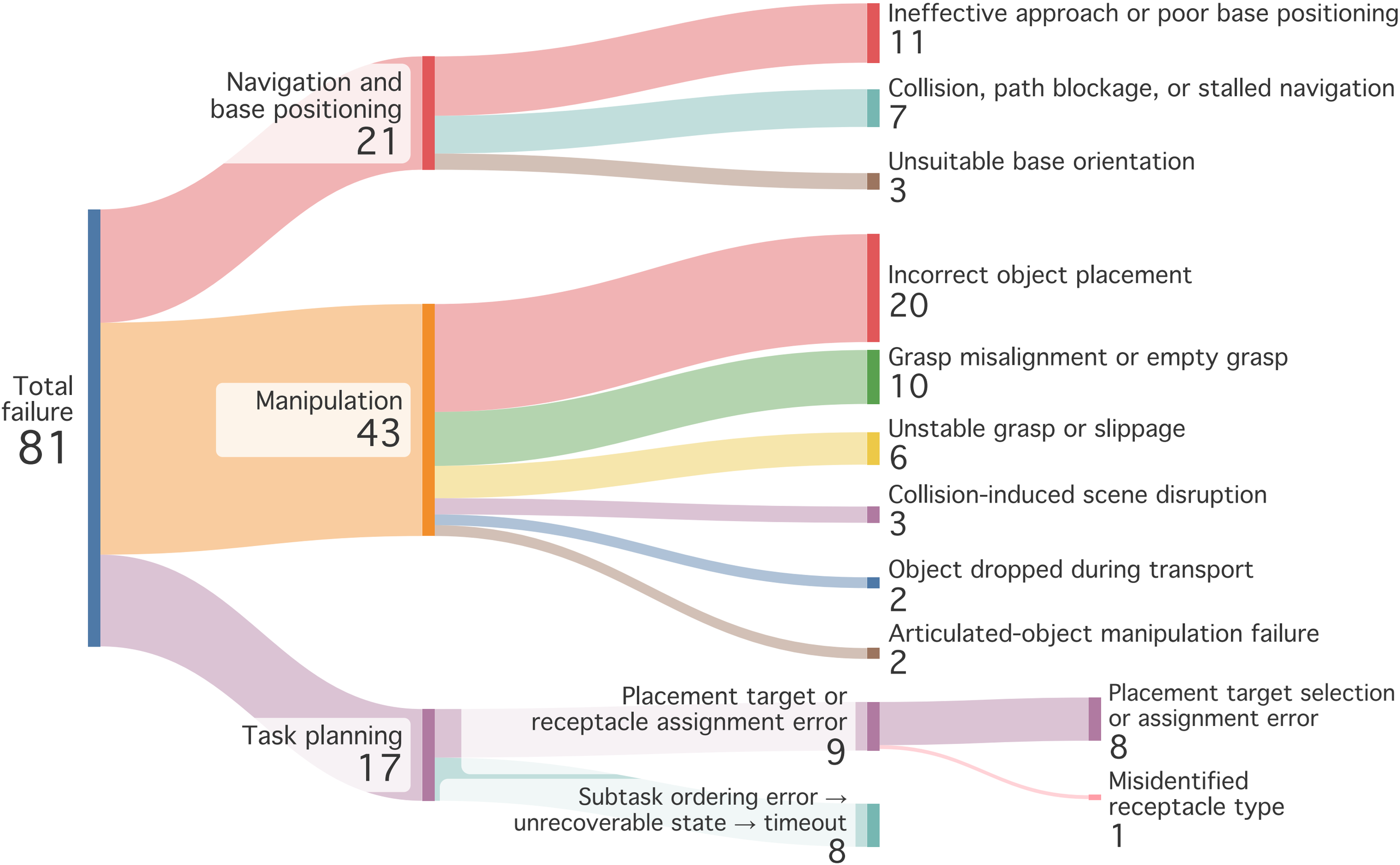
Total failure
81
Navigation and base positioning
21
Manipulation
43
Task planning
17
Ineffective approach or poor base positioning
11
Collision, path blockage, or stalled navigation
7
Unsuitable base orientation
3
Incorrect object placement
20
Grasp misalignment or empty grasp
10
Unstable grasp or slippage
6
Collision-induced scene disruption
3
Object dropped during transport
2
Articulated-object manipulation failure
2
Placement target or receptacle assignment error
9
Subtask ordering error → unrecoverable state → timeout
8
Placement target selection or assignment error
8
Misidentified receptacle type
1